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Almost 10 years of dual reporting of Scope 2: chaos or comparability?

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ABSTRACT

Investors, regulatory bodies, customers, and other stakeholders are increasingly concerned about climate change and demand meaningful, comparable and consistent corporate Greenhouse Gas (GHG) Emission Scope 1 and 2 data. Companies are responding to these demands by disclosing their annual GHG emissions and setting ambitious climate targets to reduce their impact on the environment. In 2015, the GHG Protocol has introduced a dual reporting method for estimating Scope 2 emissions and has started a review process to update its standards. Based on an in-depth analysis of corporate Scope 1 and 2 carbon disclosure from 2019 to 2023 of 102 companies in the European chemical and pharmaceutical industry, the study reveals that there exists heterogeneous disclosure practices of market- and location-based Scope 2 GHG emissions, which do not comply with the requirements set by the GHG protocol. Moreover, the analysis has identified several accounting issues and a lack of transparency when setting a reduction target and reporting progress against it. The results imply that the dual reporting method itself does not necessarily represent a problem, but the missing clear distinction of the Scope 2 methods used in corporate carbon disclosure. Based on the findings, implications for research, policy makers and standard setters are provided.

ARTICLE HISTORY

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SUBJECT

CLASSIFICATION CODES

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Introduction

In 2023, U.N. secretary general António Guterres stated that “The era of global warming has ended, and the era of global boiling has arrived” [1]. In his speech, he pointed out that reaching the Paris Agreement is still possible, but only with dramatic and immediate climate action. Companies and their emissions caused by the consumption of energy to produce goods and provide services play a major role in achieving this agreement.

To report on the state of companies’ climate target achievements and internal target tracking, but also for fulfilling investors and stakeholders’ needs and compliance with mandatory reporting regulations, companies are estimating their GHG inventories by accounting for their annual Scope 1, 2 and 3 emissions using the standard of the GHG protocol [2,3].

Furthermore, it is important to note that the corporate GHG Protocol standard has been developed separately from accounting principles of national inventory systems. The objective of the corporate GHG Protocol standard is to provide rules and accounting principles and aims at creating a

foundation for developing corporate climate strategies and provide transparency for decarbonization efforts of individual companies. Corporate GHG inventories should not be used to assess developments related to national emission reduction targets, as double-counting issues may occur [3].

Generally, the GHG protocol separates companies’ GHG inventories into three main categories. Scope 1 relates to the direct emissions from own energy generation and combustion (e.g. through combined heat and power plants), Scope 2 entails all emissions from the generation of acquired and consumed electricity, steam, heat, or cooling for the purpose of their own operations. Thus, the sum of Scope 1 and 2 entails the total operational carbon footprint of a company to produce its goods and/or provide its services. Moreover, Scope 3 refers to all GHG emissions from upstream and downstream activities to account for the whole value chain [3].

For the purposes of comparing corporate GHG emission data amongst companies, it is crucial to consider the sum of Scope 1 and 2. In the past, emphasis has been placed on Scope 1 GHG emissions, especially under the EU ETS regulations. However, from a

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This article has been corrected with minor changes. These changes do not impact the academic content of the article.

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carbon management perspective, considering only Scope 1 without Scope 2 leads to a bias in companies' performance, either for those companies focussing their energy consumption primarily on externally bought in energy (Scope 2), or for those companies focussing on own energy production (Scope 1). In corporate GHG accounting, only the sum of Scope 1 and 2 captures the full GHG emission intensity of companies' own operations. Consequently, it is very important that not only the disclosure of Scope 1 data is consistent and comparable, but also Scope 2. Upcoming regulations, including the European Sustainability Reporting Standards (ESRS) under the Corporate Sustainability Reporting Directive (CSRD), require a detailed disclosure of Scope 1 and 2 GHG emissions from both own generated and externally bought-in energy consumptions [4].

Since the publication of a specific carbon accounting Scope 2 Guidance in 2015 by the GHG protocol, companies are required to estimate and disclose two separate figures (so-called dual reporting method) for their Scope 2 GHG emissions: location-based emissions are accounted for by national or regional grid mix factors, market-based Scope 2 through pairing emission factors from different contractual instruments with the consumption of electricity consumed through the individual instrument.

Using the second method, companies might claim a zero or near-zero emission factor for electricity if provided by their individual suppliers (e.g. through 100% green electricity contracts), or associated with renewable electricity generation through dedicated renewable electricity certificates (hereafter RECs) [5]. If information on supplier emission factors is not available or there are remaining consumptions not covered by any market-based instrument, the standard requires to use a residual grid mix emission factor, which represents emissions and types of generation resources that remain after certificates, and supplier-specific factors have been claimed [6]. The following overview shows the emission factor hierarchy as defined by the GHG Protocol (see Figure 1):

The methods companies use for accounting for their Scope 2 emissions can have a significant impact on their GHG emission data and associated reported reductions. For example, in their Corporate Responsibility Report 2022, German telecommunication company Deutsche Telekom reported on both Scope 2 market-based and location-based emissions. Scope 2 emissions in 2020 accounted with the market-based method equal 21,019 tons of CO₂e, whilst location-based Scope 2 emissions equal 4,232,913 tons of CO₂e [7]. The location-

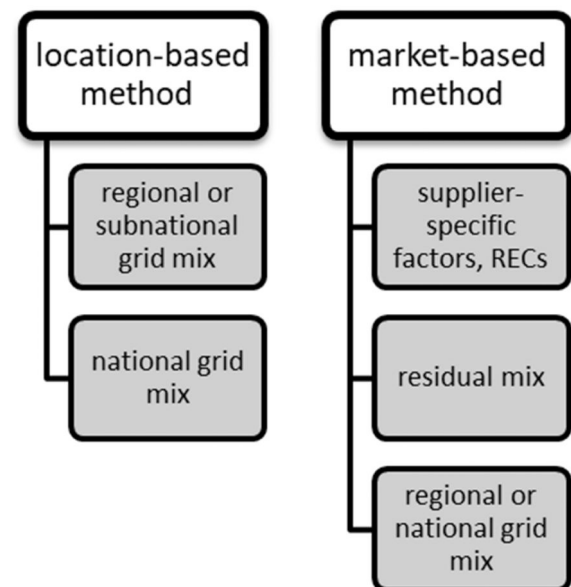


Figure 1. Scope 2 data hierarchy, based on GHG protocol guidance [5].

based Scope 2 emissions are therefore 200 times higher than the market-based Scope 2 emissions, highlighting the enormous differences depending on the method chosen.

In academia, systematic literature review studies on carbon disclosure have not addressed the dual reporting method of Scope 2 [8–11]. Qualitative, industry-specific studies on carbon disclosure and performance often provide in-depth analyses on the development of GHG emissions, but neglect different approaches of Scope 2 [12,13]. Currently, research on carbon disclosure and performance mostly ignore the differentiation between market- and location-based Scope 2 GHG emission data. For example, empirical studies on the interaction between carbon and financial performance [14], on carbon performance and disclosure [15], or on governance-related determinants and carbon disclosure [16] do not transparently explain whether the underlying Scope 2 GHG emission data is based on the market- or location-based approach. Therefore, this study sets out to explore past and current practices of Scope 1 and Scope 2 GHG emission disclosure and aims at providing insights into different Scope 2 disclosure approaches by answering the first research question: Which patterns of Scope 1 and 2 GHG emission disclosure can be identified and do companies change their reporting over time?

The GHG Protocol published the Scope 2 guidance with the intention to allow for a differentiated view on corporate Scope 2 GHG emissions. Whilst the location-based emissions provide information on the GHG emission intensity of grids where companies are operating in, the market-based GHG emissions aim at providing insights into companies' decarbonization

efforts by purchasing renewable electricity. Through the purchasing of energy (specifically electricity) from renewable sources, companies can report a significant reduction of the carbon intensity of their overall operations [5,17]. Whilst energy efficiency improvement is an important lever of reducing energy consumption, the electrification of industrial processes is one of the key pillars of global decarbonization pathways [17]. Renewable electricity from solar, wind, hydropower and other sources are essential for a rapid future transition to less carbon-intensive energy systems to limit global warming: in the IEA's net zero scenario, renewable energies are responsible for one of the largest shares of global CO₂ emission reductions [17]. As electricity becomes more and more important for companies' own production processes and switching to renewable electricity is one of the main decarbonization levers, misleading accounting of GHG Scope 2 emissions can impede global decarbonization efforts [18].

The GHG Protocol is currently in the review process of its accounting standards for Scope 2. Between November 2022 and March 2023, a comprehensive survey (400 respondents) was conducted, and proposals were collected (70 proposals of which 43 are shared publicly). The final survey results were published in November 2023 providing a comprehensive overview of different views on the topic [19]. Some respondents would like to maintain the dual reporting method, others favoured only one method to decrease confusion and double-counting issues. Supporters for location-based argue that these emissions truly represent the actual impact of companies based on physical consumption of electricity. Supporters of the market-based method argue that this method incentivizes companies to purchase renewable electricity, creating demand on the market to foster grid decarbonization [19].

Whilst there exist many different views on how a future Scope 2 accounting framework should look like, only limited research has been conducted to gain a deeper understanding of Scope 2 GHG emission accounting in practice. Some researchers argue that only location-based emissions from the grid averages reflect the GHG emissions emitted by a company through their electricity purchases [18]. Increasingly, focus of accounting research has been placed on misleading market-based accounting and the role of unbundled RECs, which have raised concern amongst organizations, policy makers and the academia related to their effectiveness and actual contribution in combatting climate change through emission reductions [6,18,20].

Currently, there exists a research gap in the carbon accounting literature that analyses whether companies not only disclose Scope 2 GHG emissions according to the dual reporting method, but also comply with the underlying accounting rules.

Therefore, this paper sets out to explore different types of Scope 2 accounting approaches and aims at answering the second research question: Are there any critical issues arising from Scope 2 practices which might be relevant for the overall functionality of the accounting system?

This is particularly important as many companies have committed to contribute to the Paris climate agreement by setting ambitious absolute Scope 1 and 2 GHG emission reduction targets [21,22]. Thus, it is crucial to provide evidence from actual cross-company data to gain insights into companies' accounting practices and analyse possible issues arising from the dual reporting method when setting and pursuing ambitious (science-based) climate targets. Therefore, this study also aims at answering the third research question:

What are the implications of the Scope 2 dual reporting method for companies' climate targets and reported emission reductions?

Sample and data collection

The selection of companies was based on companies reporting to the CDP (formerly Carbon Disclosure Project) choosing "Chemicals" and "Biotech and Pharma" as the primary sectors, leading to an initial sample of 127 companies. The chemical and pharmaceutical industries were chosen as they are one of the most important carbon-intensive sectors being responsible for one third of all responsible for a substantial share of GHG emissions in the industrial sector and providing at the same time many opportunities for decarbonization [23,24]. The European context is particularly interesting due to the upcoming reporting requirements of the CSRD, but also because there exist different countries with distinct characteristics to compare and the Guarantees of Origin (GOs) system plays an important role in companies' reporting of decarbonization efforts related to their purchased electricity [25,26]. Even though the CDP was chosen for the selection of companies, the collection of information related to corporate GHG emission disclosure was hand-collected from companies external reporting (including annual reports and/or sustainability reports). To be able to gain insights into the disclosure and accounting practices of Scope 2 and the application of the dual reporting method, in-depths

assessments of the actual reporting were required as many detailed information are not provided through external data bases. Even though several third-party providers such as Bloomberg or Thomson Reuters collect relevant ESG data, there exist issues of comparability of GHG emissions amongst them [27]. For our assessment, we considered the reporting period from 2019 to 2023 and analysed every report for every year individually. This means that the disclosure of Scope 1 and 2 reporting was assessed by not only considering the latest available report but assessing the year-on-year changes of Scope 1 and 2 disclosures. Relevant

information on climate targets were extracted from either external reporting and/or the website of the Science-based Targets Initiative. Due to missing external reporting between 2019 and 2023 or only from 2022 onwards, and due to changes in ownership structures, 25 companies were removed from the initial sample, leading to a final sample of 102 companies. Companies which published annual/sustainability reports in most years between 2019–2023 were not excluded. Generally, the final sample consists of companies which headquarters are based in different European countries, as shown in Figure 2:

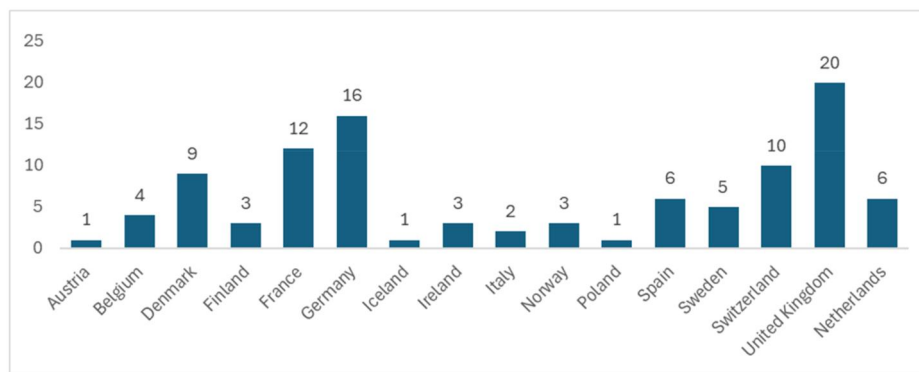


Figure 2. Number of companies per country, by headquarter ($n = 102$).

Table 1. Classification system of different disclosure categories.

Category	Description	Illustrative Example									
No report available	No report was published in the respective year	n/a									
Not published	Report was published but no Scope 1 or Scope 2 data was available	n/a									
No Scopes mentioned	GHG emission data was published, but no emission Scopes were mentioned	<table border="1"> <thead> <tr> <th></th><th>2021</th><th>2022</th></tr> </thead> <tbody> <tr> <td colspan="3">Emission control by ESM</td></tr> <tr> <td>4 CO₂e</td><td>258,095</td><td>287,526</td></tr> </tbody> </table> [28]		2021	2022	Emission control by ESM			4 CO ₂ e	258,095	287,526
	2021	2022									
Emission control by ESM											
4 CO ₂ e	258,095	287,526									
Scope 1 and 2 as a sum	GHG emission data was labelled in Scopes, but published as one figure	<table border="1"> <thead> <tr> <th></th><th>2022</th><th>2021</th></tr> </thead> <tbody> <tr> <td>GHG emissions, scope 1+2 (market based)</td><td>15.9</td><td>17.5</td></tr> </tbody> </table> [29]		2022	2021	GHG emissions, scope 1+2 (market based)	15.9	17.5			
	2022	2021									
GHG emissions, scope 1+2 (market based)	15.9	17.5									
Intensity metric published	GHG emission data was not published in absolute terms but as an intensity metric	Example 1: “SCOPE 1 GHG EMISSIONS, T CO₂eq/EMPLOYEE” [30] Example 2: “Carbon footprint scope 1 (tonnes CO₂/1,000 L paint produced)” [31]									
Scope 1 published	Scope 1 was published in report										
Scope 2 published	Scope 2 was published in report										

Results

Analysis of GHG emission disclosure and the dual reporting method

To understand current practices of GHG emission disclosure, a detailed review of Scope 1 and 2 disclosures was conducted, and distinct disclosure types were identified. Table 1 explains the different disclosure classifications and provides illustrative examples for each of the categories.

In general, many companies reported on their Scope 1 and 2 GHG emissions. However, there were distinct differences related to the metrics used and the granularity provided. In 2019, 64% of companies reported Scope 1 and Scope 2 as two separate data points. From 2019 onwards, this number constantly increased as more companies started to publish reports or included GHG emission data into their external reporting. From a disclosure quality perspective, companies enhanced their reporting: In 2019, 16% of companies did publish Scope 1 + 2 data, but reported either

intensity metrics, Scope 1 and 2 as a sum, or did not mention any Scope when publishing GHG emissions. By 2023, 86% of companies reported on separate Scope 1 and 2 figures, highlighting more harmonization and standardization across companies. Figure 3 presents the overall development between 2019 and 2023:

To assess the status quo of Scope 2 disclosure and the alignment with the dual reporting method, a more granular classification system had to be developed. During the review, many different levels of quality of disclosure, alignment with the dual reporting method, and levels of transparency could be observed. The disclosure types were collected and categorized to allow for a meaningful analysis of individual cases. In total, ten different types of Scope 2 disclosure were identified. Table 2 provides an overview of all classifications.

In 2019, four years after the publication of the dual reporting method guidance for Scope 2 GHG emissions, only 23% of companies published both market- and location-based Scope 2 data. 17% of

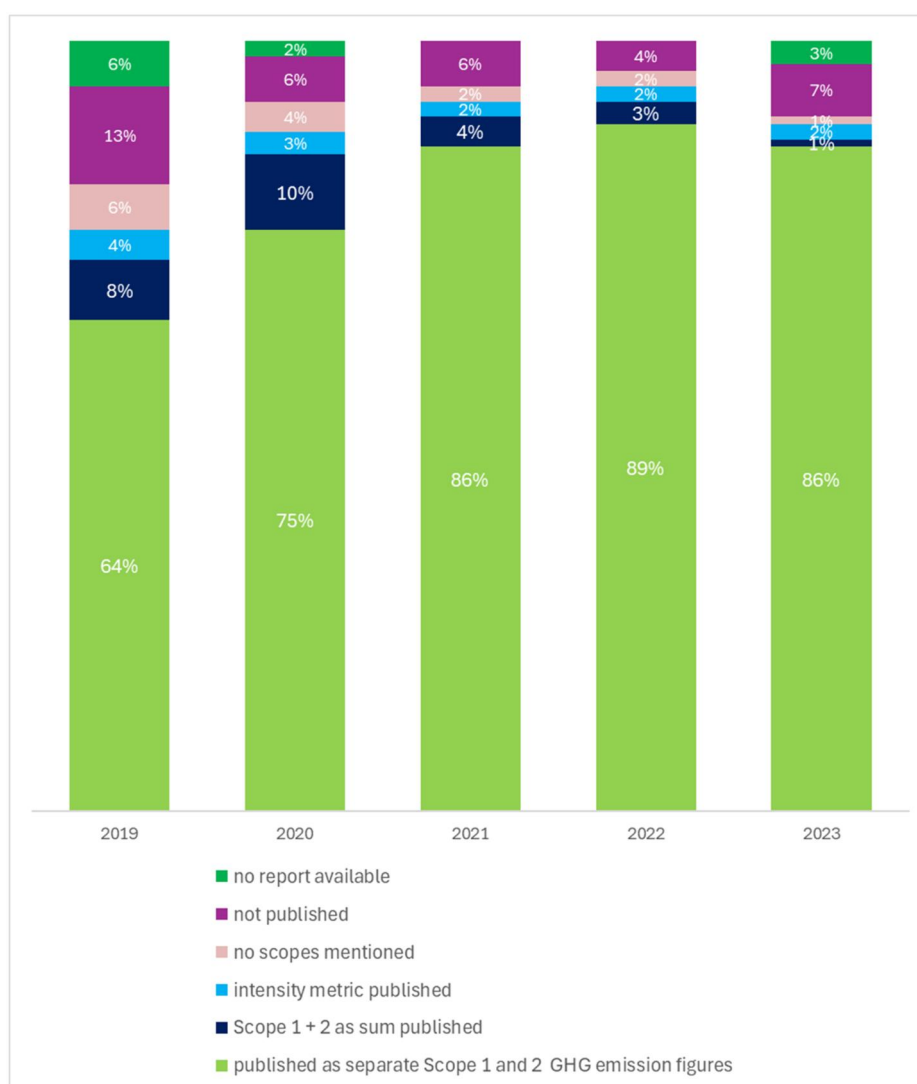


Figure 3. Shares of Scope 1 and Scope 2 disclosure over time ($n = 105$).

Table 2. Classification system of Scope 2 disclosure categories.

Category	Description	Illustrative Example			
market- and location-based published: equal hierarchy	Report gives equal importance to the two emission figures; can be found in the same section	Description of the indicator		2023	2022
		Carbon Scope 2 Total Emissions (tCO ₂ E) <i>Location-based methodology</i>		6,588	6,890
		Carbon Scope 2 Total Emissions (tCO ₂ E) <i>Market-based methodology</i>		367	1,567
[32]					
no labelling market- or location-based	Scope 2 data is not labelled and there exists no explanation with method was used for calculating data			2023	2022
		Scope 1 and 2 GHG emissions			
		GHG emissions (Scope 2)	metric tonnes of CO ₂ e	344,277	484,839
[33]					
market- and location-based published: location-based in footnote	Market-based Scope 2 data is highlighted, location-based data is presented in footnote only			2023	2022
		Absolute emissions (million tons CO₂ eq.)			
		Direct emissions, scope 1	1.00	0.92	
		Indirect emissions, scope 2	0.34	0.35	
		Total scope 1 & 2 GHG emissions ^{b)}	1.34	1.27	
^{b)} Scope 2 emissions according to the location-based approach are 0.64 mn tons CO ₂ equivalents in 2023.Scope 2 emissions according to the location-based approach are 0.64 mn tons CO ₂ equivalents in 2023					
[34]					
market- and location-based published: location-based in GRI Index	Market-based Scope 2 is highlighted in main report, location-based data can be found in appendix/GRI index only –			2023	2022
		Scope 2 emissions (t CO ₂ e) ¹	1,552	1,322	
		¹ The Scope 2 emissions reported in this section are based on the market-based method.			
[35]					

(continued)

	sometimes in different documents	<table><tr><th colspan="2">PCs</th><th>Offices</th><th>Unit</th></tr><tr><td>Scope 2 – GHG emissions (location-based) - Total</td><td>43,820.26</td><td>6,592.77</td><td>tCO2e</td></tr></table> [36]	PCs		Offices	Unit	Scope 2 – GHG emissions (location-based) - Total	43,820.26	6,592.77	tCO2e										
PCs		Offices	Unit																	
Scope 2 – GHG emissions (location-based) - Total	43,820.26	6,592.77	tCO2e																	
market- and location-based published: location-based in text	Market-based Scope 2 is presented prominently in table; location-based data can be found in text only	“Scope 2 emissions were calculated using the ‘market-based’ approach in accordance with the GHG Protocol Scope 2 Guidance. When reported according to the ‘location-based’ approach, the Scope 2 emissions were 17,712 tCOe.” [37]																		
market- and location-based published: location-based as sum s1+2	Market-based data is presented separately, location-based data only as sum	<table><tr><th colspan="2">2023</th><th>2022</th></tr><tr><td colspan="3">Corporate Carbon Footprint by Emissions Category (in tCO2e)</td></tr><tr><td>Scope 1: Direct emissions</td><td>13,375</td><td>13,908</td></tr><tr><td>Scope 2: Indirect emissions</td><td>3,930</td><td>6,553</td></tr><tr><td>Total Scope 1 and 2 (market based)</td><td>17,305</td><td>20,461</td></tr><tr><td>GHG emissions Scope 1 and 2 (in tCO2; location-based)</td><td>32,881</td><td>31,622</td></tr></table> [38]	2023		2022	Corporate Carbon Footprint by Emissions Category (in tCO2e)			Scope 1: Direct emissions	13,375	13,908	Scope 2: Indirect emissions	3,930	6,553	Total Scope 1 and 2 (market based)	17,305	20,461	GHG emissions Scope 1 and 2 (in tCO2; location-based)	32,881	31,622
2023		2022																		
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Total Scope 1 and 2 (market based)	17,305	20,461																		
GHG emissions Scope 1 and 2 (in tCO2; location-based)	32,881	31,622																		
market- and location-based published: market-based in footnote	Location-based Scope 2 data is highlighted, market-based data is presented in footnote only	<table><tr><th colspan="2">2023</th><th>2022</th></tr><tr><td>Scope 2, location based (tCO2 e) – Electricity^{6,8}</td><td>13,663</td><td>13,573</td></tr></table> ⁸ In 2023, ICON calculated market-based Scope 2 emissions for the first time which were verified by Carbon Trust, these emissions were 4,969.81 tCO2e. [39]	2023		2022	Scope 2, location based (tCO2 e) – Electricity ^{6,8}	13,663	13,573												
2023		2022																		
Scope 2, location based (tCO2 e) – Electricity ^{6,8}	13,663	13,573																		
only market-based published	No data on location-based Scope 2 emission available	<table><tr><th colspan="2">2023</th><th>2022</th></tr><tr><td>Greenhouse gas emissions Scope 1</td><td>290,424</td><td>308,925</td></tr><tr><td>Greenhouse gas emissions Scope 2*</td><td>0</td><td>0</td></tr><tr><td>Greenhouse gas emissions Scope 3</td><td>1,846,871</td><td>1,892,528</td></tr></table> *The significant decrease since 2020 in comparison with previous years is due to the	2023		2022	Greenhouse gas emissions Scope 1	290,424	308,925	Greenhouse gas emissions Scope 2*	0	0	Greenhouse gas emissions Scope 3	1,846,871	1,892,528						
2023		2022																		
Greenhouse gas emissions Scope 1	290,424	308,925																		
Greenhouse gas emissions Scope 2*	0	0																		
Greenhouse gas emissions Scope 3	1,846,871	1,892,528																		

(continued)

		fact that Symrise has demonstrably sourced its global external electricity from renewable sources and offset the CO₂ emissions resulting from the use of steam through climate protection projects. This means that the volume of Scope 2 GHG emissions could be reduced to zero in 2023. <table border="1"> <thead> <tr> <th></th><th>2023</th><th>2022</th></tr> </thead> <tbody> <tr> <td>Total GHG emissions (market-based) (tCO₂eq)*</td><td>2,137,295</td><td>2,201,453</td></tr> </tbody> </table> *Symrise currently only reports its greenhouse gas emissions using a market-based method [40]		2023	2022	Total GHG emissions (market-based) (tCO ₂ eq)*	2,137,295	2,201,453
	2023	2022						
Total GHG emissions (market-based) (tCO ₂ eq)*	2,137,295	2,201,453						
only location-based published	No data on market-based Scope 2 emission available	<table border="1"> <thead> <tr> <th></th><th>2023</th><th>2022</th></tr> </thead> <tbody> <tr> <td>Scope 2: Energy indirect GHG emissions, location based (Base year 2020) t CO₂ e</td><td>64,093</td><td>56,511</td></tr> </tbody> </table> [41]		2023	2022	Scope 2: Energy indirect GHG emissions, location based (Base year 2020) t CO ₂ e	64,093	56,511
	2023	2022						
Scope 2: Energy indirect GHG emissions, location based (Base year 2020) t CO ₂ e	64,093	56,511						
Not understandable	GHG emission data was published but estimation of market-based and location-based Scope 2 emission was not understandable	“Since fiscal year 2018/19 ZEISS has been collecting data on both location-based Scope 1 and Scope 2 emissions. Market-based emissions are calculated using specific emission factors relating to the energy providers. The site-based survey uses emission factors from the respective federal state where energy is consumed. For this purpose, ZEISS uses the current emission factors according to the VDA list (as of 2020, published in December 2022). Market-based emissions have been reported since fiscal year 2020/21. The site-based Scope 1 and Scope 2 emissions amounted to 252.012 t CO₂-eq in FY 2022/23, 243,538 t CO₂-eq in FY 2021/22, 265,737 t CO₂-eq in FY 2020/21, 237,308 t CO₂-eq in FY 2019/20 and 236,052 t CO₂-eq in FY 2018/19” [42]						

companies give equal importance to the two Scope 2 GHG emission figures in their reporting, whilst the remaining 6% report location-based Scope 2 in different locations than market-based Scope 2. In 2019, most companies do not label their Scope 2 emissions (32%) as either market-based or location-based. Consequently, there exists a gap as external readers have no information on the underlying accounting method chosen. Moreover, 5% of companies published only market-based Scope 2 data; and 3% only location-based Scope 2 data; leading

to a confusing situation where it is hard to conduct meaningful comparisons amongst companies. [Figure 4](#) provides an overview of the development over time and the shares of disclosure types in the respective years:

By 2023, the picture changes significantly: Within five years, the number of companies reporting both market- and location-based Scope 2 rose to 62%. Whilst these developments are generally showing more homogeneity amongst companies' Scope 2 disclosure, in 2023 49% of companies do

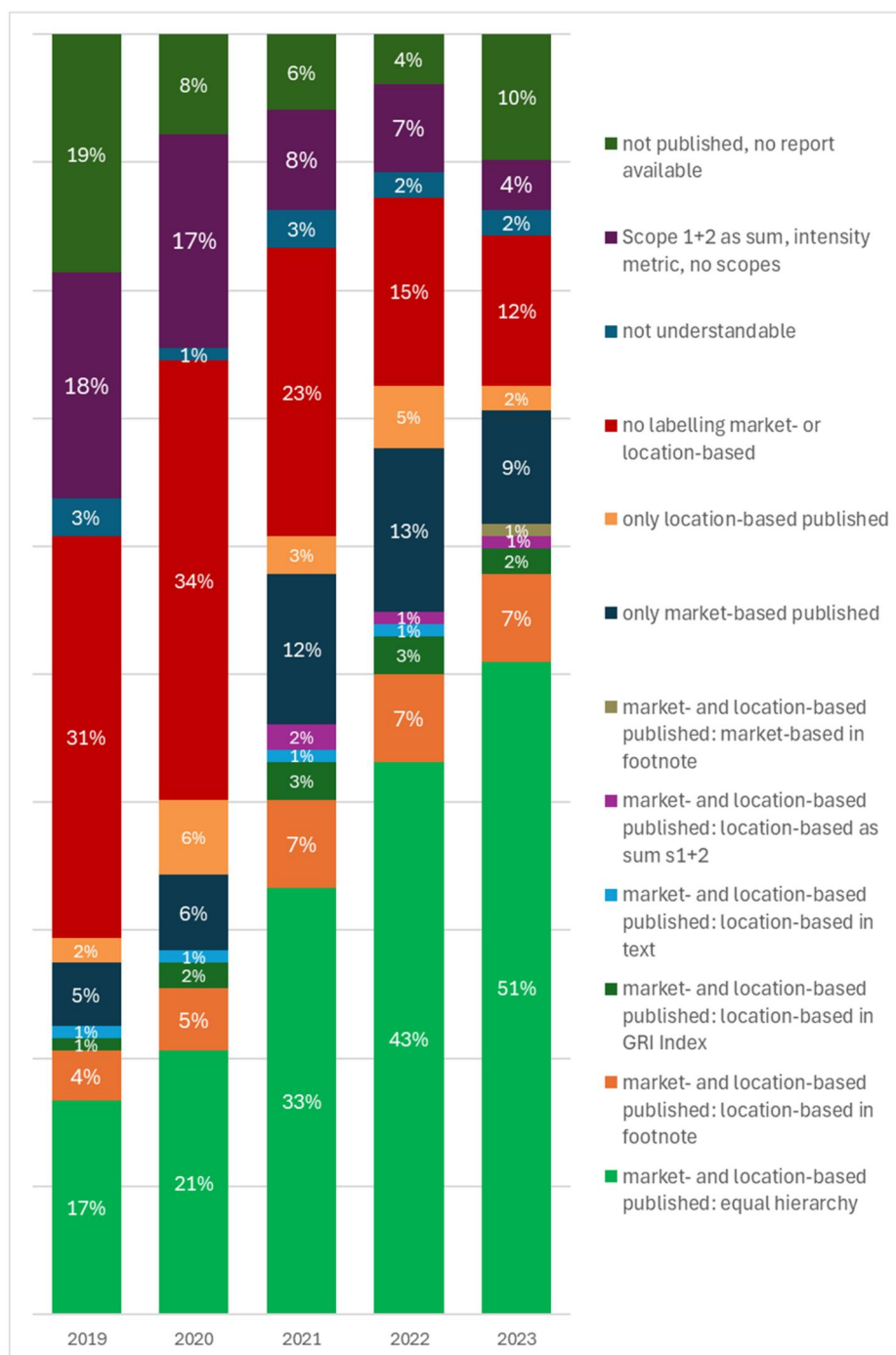


Figure 4. Shares of Scope 2 disclosure types over time ($n = 105$).

not report in line with the dual reporting requirements of Scope 2, as intended by the GHG Protocol. In 2023, 23% of companies either do not label Scope 2, or only publish market- or location-based data. In contrast to Scope 1, the complexity of accounting for bought-in electricity appears to be more challenging to the intended users of the standard. This, in turn, poses an enormous difficulty for the appropriate necessary comparability amongst companies.

Moreover, compared to 2019, more companies decided to report on a more individual level, as 7% of companies reported location-based data separately in a footnote. 10% of companies reported

market-based data more prominently in a table, and location-based data in a footnote (7%), in an Appendix/GRI¹ Index (2%), or only as a sum of Scope 1 and 2 (1%). In contrast, only one company decided to report location-based data in a table, and market-based Scope 2 in a footnote. Moreover, the number of companies publishing only market-based Scope 2 data increased from 2019 (5%) to 2023 (9%). Consequently, it seems likely that companies are putting more emphasis on market-based figures, whilst location-based seems to be neglected in many cases. Moreover, there exist specific accounting issues related to Scope 2, especially in market-based accounting, which add to the issue of comparability

amongst companies, and which will be further investigated on in the subsequent chapter.

Accounting issues identified for Scope 2

To investigate not only on the types of disclosure, but also on different types of accounting approaches, the accounting methodologies were screened to analyse the compliance with the rules set by the GHG Protocol. In general, two accounting issues were identified. The first one relates to mixing up methods and a missing application of the residual mix in market-based Scope 2 accounting. The second one relates to the temporal matching of the consumption of renewable electricity and the usage of zero/near-zero emission factors in Scope 2 market-based accounting.

The GHG Protocol Scope 2 guidance highlights that the two accounting methods should always be considered separately from each other. A clear differentiation between market-based and location-based data and the application of the market-based emission factor hierarchy is important due to issues related to possible double-counting of renewable electricity fed into the grid. Especially in Europe and North America, which both have implemented a cross-country system of selling renewable electricity certificates, it is crucial to ensure that the two methods are not mixed up and renewable electricity is not double counted [26].

The example of Norway explains possible implications of mixing up Scope 2 accounting methods [26]. Many countries have a long history of renewable electricity production, such as Norway. Using the location-based method, companies operating in Norway can claim an emission factor close to zero as renewable sources in the grid account for around 98% [43]. However, Norway is also the largest issuer of GOs, having exported around 90 TWh of GOs in 2023 [44]. These GO certificates are typically purchased by other electricity providers and sold to European companies and (after being accounted for in Scope 2), redeemed in the respective country registries. Consequently, when extracting the sales of renewable electricity (certificates) in the market to other European countries, Norway's residual mix factor equals 599 gCO₂/kWh [45].

Thus, whenever companies in Norway use the market-based method for Scope 2 accounting, they would have to use the residual mix to account if any amounts that are not matched by renewable electricity instrument(s) [6]. In the sample, there

exist many cases where companies applying the market-based accounting but without integrating the residual mix factors, for example (Table 3):

This constitutes an issue as companies are not complying with the emission factor hierarchy of the GHG Protocol and underestimating market-based Scope 2 GHG emissions in the respective years, as the residual mix factor is (on average) higher than the grid mix.

Moreover, a few companies seem to have mixed up the Scope 2 accounting methods. For example, in their 2022 annual report, Treated PLC states the following: "We continue to use 100% renewable electricity in the UK, playing our part in stimulating growth in the renewable energy market. This year we've increased our accuracy with a more rigorous approach by including location-based Scope 2 emissions for this renewable electricity usage in the UK, in our global emissions" [49].

Using 100% renewable electricity should be reflected in market-based accounting, and not in location-based accounting as the latter is based on grid mixes.

Table 3. Description of market-based accounting approaches without or not fully understandable consideration of residual mix factors.

Exemplary market-based accounting approaches	
[46]	"All Scope 2 greenhouse gases are calculated as CO ₂ e. In 2020, the conversion factors used were provided by the energy producers. Where these were not available, factors from the IEA (International Energy Agency) from 2018 were used for fiscal year 2020, factors from 2017 were used for fiscal year 2019 and factors from 2016 were used for fiscal year 2018. The Scope 2 CO ₂ e emissions are calculated using the market-based method in accordance with the GHG Protocol."
[47]	"Scope 2: the main emissions from operated sites are related to electricity, for which consumption data is taken from supplier invoices. The emission factors used come directly from the sites' electricity suppliers, using the Market-based approach, or from the International Energy Agency (IEA) if the data are not available."
[48]	"Hybrid Approach: using location base info except for sites within the Group that purchase certified "Green" electricity. Electricity for these locations has been given a CO ₂ e emission factor of zero in calculating energy-related emissions totals. These include the sites in the Netherlands, Spain, Marl (Germany) and all sites in the UK. The hybrid approach is the approach that has been used by the Group to establish the baseline and the targets."

Table 4. Location-based Emissions reported by Tecan Group [50].

Location-based approach			
Scope	GHG category	2023 (tCO ₂ e)	2022 (tCO ₂ e)
Scope 2		7,178.16	7,178.12
Scope 2	Electricity: Grid	3,768.84	6,019.33
Scope 2	Electricity: Renewable	3,330.43	1,065.39
Scope 2	Electricity: Electric vehicles	3.72	–
Scope 2	Cooling	1.07	–
Scope 2	Heating	74.10	93.40

The scope 2 bold figures represent the sums of the data below.

Another example can be found in the report of Tecan Group [50]:

In this table, location-based approach differentiates between electricity from the grid and from renewable. However, this should be included in market-based accounting as the location-based estimation is based on regional or country-specific grid mixes.

Another critical aspect for accurate Scope 2 accounting relates to the difference between the point of time where the renewable electricity is fed into the grid and the use of the associated credits in Scope 2 market-based accounting. It is important that one kWh of renewable electricity fed into the grid should only be accounted for once to derive zero or near-zero emission factors for Scope 2. For an accurate accounting, companies must assess the amount of renewable electricity purchased on an annual basis, as different contracts with varying shares of renewable consumptions might occur. The case of Guerbet raises an issue as a constant emission factor is used for market-based Scope 2 emissions: “To reflect Guerbet’s performance without taking exogenous decarbonization factors into account, we take green energies into account as part of the reduction and calculate emissions related to energy consumption with a constant emission factor. In other words, we take into account the emission factors used in 2021: the emission factors used are those from 2015 for gas and 2020 market-based for electricity (market-based means taking into account the electricity suppliers chosen by the companies)” [51].

These examples raise the question whether market- and location-based Scope 2 estimations challenge many companies as some important accounting mistakes have been identified. Especially market-based accounting seems to be a challenge, which constitutes an issue for the overall accuracy of emission allocation in the underlying book-and-claim energy system and comparability of carbon performance amongst companies.

Scope 2 and climate targets

Generally, in line with GHG Protocol, but also target standard setter such as the science-based targets initiative (SBTi), companies can either choose between location-based or market-based accounting for defining and tracking Scope 1 and 2 GHG emission targets.

To elaborate on the research question: “What are the implications of the Scope 2 dual reporting method for companies’ climate targets and reported emission reductions?”, the current

corporate climate targets were screened to assess whether companies are transparent about their Scope 2 accounting method chosen for target tracking. In general, out of the 102 companies in the final sample, 84 companies have set absolute (near-term) targets, of which 61 were validated by the SBTi. Generally, the analysis shows that companies are vague about the method chosen for achieving absolute reduction Scope 1 and 2 targets. 44% of the companies did not explain the underlying Scope 2 method for their climate target tracking. As companies either do not label their Scope 2 data at all or published both figures, it was not possible to fully link developments of Scope 1 and 2 GHG emissions and the target achievements in the respective reporting years. Out of all 84 companies with an absolute GHG emission reduction target, 50% labelled their target to be achieved using the market-based method in their external reporting.

This reflects the intention of the market-based method to allow for transparency on individual corporate climate action and enabling stakeholders to compare and recognize different progress of decarbonization amongst companies [5].

Nevertheless, five companies in the sample decided to use the location-based method for Scope 2 target tracking. The fact that different methods chosen for climate targets is interesting as, in the current system, double-accounting of renewable electricity is common: Using the earlier example, if the user of the GO outside Norway uses the market-based method for Scope 2 accounting, and the Norwegian company uses the location-based method, a single renewable electricity attribute would be used and reported as an absolute emission reduction by both companies.

Despite that, as companies move towards dual reporting or labelling Scope 2 as being based on either market- or location-based, irregularities were identified related to the consistent use of the same methodology over time. This means that, in some cases, companies disclosed Scope 1 and 2 in the base year in a different way than in the subsequent years. For example, after setting an absolute reduction target, Scope 2 was labelled as market- or location based in the subsequent external reporting, but the emissions from the base year were not restated and not labelled as being market- or location-based Scope 2.

For example, Stratec SE has set a science-based target and has committed to the SBTi in 2023. 2019 was chosen as base year for target setting. In 2020 and 2021, Scope 2 was reported without labelling a

method for Scope 2. This changed in reporting 2022, when Scope 2 was labelled as market-based and location-based data was included in a footnote.

Figure 5 provides an overview of Scope 1 and 2 Emission disclosure between 2019 and 2023.

In the 2023 report, progress related to the science-based target is reported. The development shows progress related to the development of Scope 1 and 2 emissions between 2019 and 2023, which are mainly attributed to the significant reduction of Scope 2 between 2019 and 2020.

However, in the reporting years between 2019 and 2021, the company did not label their Scope 2 emissions, and there was no subsequent clarification which method was used in the given year. Consequently, there is no transparency about whether 2019 was also based on market-based emission factors, or whether this calculation was based on location-based grid mixes. It should be

noted that this is only an example of many throughout the sample. There exist many companies which did not label their base years adequately as a clear differentiation between market- and location-based was introduced at a later point in time. Thus, external readers do not have full transparency about the methods chosen and whether the reported GHG emission reductions are based on actual mitigation measures or changes in Scope 2 accounting methodologies.

Discussion, implications and limitations

Considering the increasing focus of the civil society and investors on corporate climate data, and many companies setting ambitious climate targets which are considered “science-based”, there exists a crucial need to harmonize and enhance companies’ disclosure and accounting practices of Scope 2

2020: No differentiation market- or location-based:			
Scope 1 and scope 2 emissions in tonnes (CO ₂ Equivalents)			
	2020	2019	Change
Scope 1			
Combustion at stationary plants	578.9	569.8	+1.6%
Combustion at mobile plants	214.2	290.7	-26.3%
Scope 2			
Electricity consumption	20.6	1,288.1	-98.4%
Total Scope 1+2	813.7	2,148.6	-63.8%
Stratec SE, annual report 2020			

2021: No differentiation market- or location-based:			
Scope 1 and scope 2 emissions in tonnes (CO ₂ Equivalents)			
	2021	2020	Change
Scope 1			
Combustion at stationary plants	581.1	578.9	+0.4%
Combustion at mobile plants	181.9	214.2	-15.1%
Scope 2			
Electricity consumption	31.0	20.6	+50.5%
Total Scope 1+2	794.0	813.7	- 7.6%
Stratec SE, annual report 2021			

2022: Scope 2 labelled as market-based, location-based figure in footnote:			
Scope 1 and scope 2 emissions in tonnes (CO ₂ Equivalents)			
	2022	2021	Change
Scope 1			
Combustion at stationary plants	459.3	502.1	-8.5%
Combustion at mobile plants	207.4	181.9	+14.0%
Scope 2 (market-based)			
Electricity consumption	31.7	31.0	+2.3%
Total Scope 1+2	698.4	715.0	- 5.3%
Stratec SE, annual report 2022			

2023: Scope 2 labelled as market-based, location-based figure in footnote:			
Scope 1 and scope 2 emissions in tonnes (CO ₂ Equivalents)			
	2023	2022	Change
Scope 1			
Combustion at stationary plants	394.7	459.3	-14.1%
Combustion at mobile plants	244.4	207.4	+17.8%
Scope 2 (market-based)			
Electricity consumption	284.3	31.7	+796.8%
Total Scope 1+2	923.4	698.4	+24.5%
Stratec SE, annual report 2023			

Figure 5. Development of Scope 1 and 2 emission disclosure and target achievement of Stratec SE [52–56].

and transparency related to science-based target achievements. The in-depth analysis of 102 companies in the European chemical and pharmaceutical industry has shown that consistent and comparable Scope 2 data and comparable target tracking is not provided in the current system.

From the perspective of carbon disclosure, several implications for research, standard setters and regulators, as well as practitioners can be derived. In 2019, most companies in the sample did not comply with the dual reporting method for Scope 2, either by only publishing one figure, or by not making transparent which method was chosen for estimating the reported data. Only recently, companies have become more aware of the distinction between market- and location-based accounting. However, there still exist a variety of different disclosure types and levels of transparency provided by companies. Currently, there exist new research avenues related to Scope 2 and the dual reporting method. Based on the results of this study, it might be valuable to understand which factors and determinants contributed to the increasing quality of Scope 2 disclosure over time; to understand why these differences occur and which linkages between Scope 2 disclosure and Scope 2 performance exist. By integrating market- and location-based Scope 2 into research, more insights into companies' carbon disclosure and performance strategies might be obtained.

From a standard setting and regulatory perspective, this study has highlighted that existing reporting regulations have not fulfilled stakeholders' needs for comparable Scope 1 and 2 GHG emission data. In 2019, a great share of companies in the sample has been publishing voluntary GRI-aligned reports. At that time, the GRI 305 Emission Standard (2016) states that location-based Scope 2 shall be reported; and, if applicable, also market-based Scope 2 emissions [57]. Only 25% of companies did publish location-based data in 2019, which rises the question why these inconsistencies were not detected, as many of these reports were externally audited by third parties.

Also, the Non-financial Reporting Directive (NFRD), which had already been in place for two years in 2019 targeting large European companies to report on material non-financial topics, seems to suffer from deficiencies as no granular reporting requirements related to Scope 1 and 2 were provided. With the new CSRD requirements and the comprehensive ESRS Standards, companies will

have to report much more granular on GHG emission figures in a separate table.

However, on a global level, there are major inconsistencies related to how Scope 2 disclosure will develop in future. Whilst all important sustainability reporting standards and regulation on sustainability/climate disclosure refer to the GHG Protocol as an underlying accounting standard for measuring corporate GHG emissions, the ESRS standards require both market- and location-based, the IFRS S2 Climate Standard requires to publish location-based Scope 2 data and optional market-based; and the Securities and Exchange Commission (SEC) Climate Disclosure Rule allows to choose one of the methods for external reporting [58,59]. Given the fact that the maturity of non-financial reporting is very diverse around the globe, the GHG Protocol could strengthen its engagement with these organisations and governments to support harmonization by creating common (dual reporting) rules for Scope 2 disclosure. For practitioners, this study highlights that being compliant with the dual reporting Scope 2 method prepares for future reporting requirements (independent of the standard chosen), whilst at the same time fulfils stakeholder needs as both Scope 2 figures are valuable for different purposes.

Whilst the impact of the GHG Protocol is rather limited on enabling standardization of corporate carbon disclosures, there exist important implications for the updates of its accounting standards based on this study's findings. Generally, the disclosure of two different figures allows for a differentiated view: firstly, location-based data represents the GHG data from an inventory point of view, based on the average grid mixes of the countries where companies are operating in. In contrast, market-based emissions enable companies to prove individual decarbonization efforts by opting for renewable electricity instruments.

This study has shown that the dual-reporting method itself does not necessarily represent a problem, but the missing clear distinction of the underlying Scope 2 accounting methods and the heterogeneous disclosure practices of companies. Whilst every method has a reason to exist, non-compliance with the standard and incorrect application of accounting requirements leads to double counting of renewable electricity fed into the grid and deficiencies in comparing companies' carbon performance. The study supports the criticism of survey respondents that the current standard is confusing and might be too complex to comply

with. As the results have shown, Scope 2 appears to be more challenging for companies to estimate in two distinct ways. Hence, the GHG Protocol might also consider effective ways of supporting companies' implementation to decrease complexity and enable a functioning overall accounting system. This also entails more educational material, learning sessions, best-practice case studies or other forms of engagement for creating a common understanding amongst the intended users of the standard. It also means that the GHG Protocol should place a greater emphasis on clarifying the primary objectives of the individual Scope 2 data.

In future standard setting, especially for Scope 2, the new GHG Protocol standard setting process should also acknowledge the fact that many of (new) sustainability reporting standards will not be changed for some time. For example, the new ESRS standards will not be updated in the next years and will therefore still require market- and location-based Scope 2 data independent of the GHG Protocol standard developments [4]. This means that, if the dual reporting method would be abolished, European companies would face a situation where Scope 2 might have to be calculated based on two different GHG Protocol standards market-based and location-based figures for reporting under the CSRD and applying new rules for Scope 2 accounting in line with the updated GHG Protocol standard. If the GHG Protocol changes from dual reporting to one single accounting rule, there exists the risk of a further lack of consistent accounting methods used and, consequently, comparability over time and across regions (as different standards and reporting regulations are in place).

Consequently, the GHG Protocol should keep the dual reporting method to cover all existing circumstances and focus on enhancing the rules for existing accounting methods based on their individual objectives. For location-based accounting, more efforts should be placed on a more granular approach than using regional or national grid mixes, leading to meaningful GHG inventories from an energy consumption perspective. For market-based accounting, the criteria of additionality of renewable electricity will be key in highlighting the efforts of companies to mitigate GHG emissions through their purchasing decisions. Moreover, the market-based method only makes sense if a functioning "book-and-claim" system is in place. If a company uses contractual purchase of zero-emission electricity to claim a low emission factor for

their electricity use, it must be ensured that in market-based accounting, this renewable electricity is not claimed twice.

Based on the results of the sample, many research avenues in the field of carbon accounting can be derived. For example, more research is needed on Scope 2 carbon accounting practices and possible issues to understand impacts on double-counting of renewable electricity, including implications on global mitigation efforts.

The study also provides insights to the question on whether there exist any Scope 2 accounting issues related to methodological rigour and consistency over time. In this context, examples were provided to illustrate a lack of transparency regarding the base year Scope 2 estimates and possible Scope 2 method changes over time. The results call for a higher emphasis on transparency and an annual reporting of an (updated) base year, to ensure methodological consistency. In this context, companies pursuing an ambitious climate strategy should always use the same Scope 2 method for target tracking and implement a recalculation policy to ensure consistency and comparability over time.

Lastly, this study assessed possible issues arising from the dual reporting method on reported climate targets. Generally, the study has provided insights into the choice of Scope 2 methods for corporate climate targets, highlighting that there exists a lack of transparency and a tendency to use market-based figures to track progress.

Future research might further investigate on why companies choose either market- or location-based Scope 2 and implications of missing defined Scope 2 data, for example if remuneration is based on the reduction of Scope 1 and 2. In this context, standard setters such as the SBTi should provide information on the Scope 2 method chosen by the individual companies on their overview of companies taking action [22]. The options to choose different Scope 2 accounting methods for science-based climate targets and possible implications for double-counting, especially on European level, represent another interesting field for further research. Moreover, as transition plans become more common, standard setter should also consider Scope 2 accounting by assessing the role of market-based instruments as long-term decarbonization levers.

There are a few limitations relevant to this study. First, the European context is specific due to cross-country mechanisms and the trade of GOs across borders, which might have led to certain approaches

of companies on how to account for Scope 2 and what to disclose. In different regions around the world, the development of Scope 2 disclosure and compliance with the dual reporting method might look completely different, also because European companies have faced increasing pressure to act from various stakeholders over the past years. Moreover, the chemical and pharmaceutical industry is not representative for all industries: There exist industries which rely less on electricity consumption, e.g. in the energy or maritime sector. Hence, disclosure quality might vary across industries depending on the importance of Scope 2 for overall decarbonization. However, this study highlights important pressuring issues which can be generalized to derive concrete recommendations for future standard setting.

Note

1. GRI: Global Reporting Initiative.

Disclosure statement

The authors report there are no competing interests to declare.

Data availability statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

References

1. United Nations. Hottest July ever signals 'era of global boiling has arrived' says UN chief; 2023. [Online]. Available from: <https://news.un.org/en/story/2023/07/1139162>.
2. Harangozo G, Szigeti C. Corporate carbon footprint analysis in practice – with a special focus on validity and reliability issues. *J Cleaner Prod.* 2017;167:1177–1183. doi: [10.1016/j.jclepro.2017.07.237](https://doi.org/10.1016/j.jclepro.2017.07.237).
3. WBCSD and WRI. The greenhouse gas protocol standard. Washington, D.C., USA: WBCSD and WRI; 2004.
4. European Commission. The Commission adopts the European Sustainability Reporting Standards; 2023. [Online]. Available from: https://finance.ec.europa.eu/news/commission-adopts-european-sustainability-reporting-standards-2023-07-31_en.
5. WBCSD and WRI. Scope 2 guidance. Washington, D.C., USA: WBCSD and WRI; 2015.
6. Bjørn A, Lloyd SM, Brander M, et al. Renewable energy certificates threaten the integrity of corporate science-based targets. *Nat Clim Chang.* 2022;12(6):539–546. doi: [10.1038/s41558-022-01379-5](https://doi.org/10.1038/s41558-022-01379-5).
7. Deutsche Telekom. Corporate Responsibility Report 2022 2022. [Online]. Available from: <https://www.cr-report.telekom.com/2022/#0>
8. Velte P, Stawinoga M, Lueg R. Carbon performance and disclosure: a systematic review of governance-related determinants and financial consequences. *J Cleaner Prod.* 2020;254:120063. doi: [10.1016/j.jclepro.2020.120063](https://doi.org/10.1016/j.jclepro.2020.120063).
9. Hahn R, Reimsbach D, Schiemann F. Organizations, climate change, and transparency: reviewing the literature on carbon disclosure. *Organ Environ.* 2015;28(1):80–102. doi: [10.1177/1086026615575542](https://doi.org/10.1177/1086026615575542).
10. Ali W, Wilson J, Frynas JG. Corporate governance mechanisms and carbon disclosure: a multilevel and multitheory literature survey. *Corp Soc Responsibility Env.* 2024;31(6):5670–5689. doi: [10.1002/csr.2869](https://doi.org/10.1002/csr.2869).
11. Borghei Z. Carbon disclosure: a systematic literature review. *Account Fin.* 2021;volume61(4):5255–5280. doi: [10.1111/acfi.12757](https://doi.org/10.1111/acfi.12757).
12. Belkhir L, Elmeligi A. Carbon footprint of the global pharmaceutical industry and relative impact of its major players. *J Cleaner Prod.* 2019;214:185–194. vol-umedoi: [10.1016/j.jclepro.2018.11.204](https://doi.org/10.1016/j.jclepro.2018.11.204).
13. Boiral O, Brotherton M, Rivaud L, et al. Comparing the uncomparable? An investigation of car manufacturers' climate performance. *Bus Strat Env.* 2022;31(5):2213–2229. doi: [10.1002/bse.3017](https://doi.org/10.1002/bse.3017).
14. Lewandowski S. Corporate carbon and financial performance: the role of emission reductions. *Bus Strat Env.* 2017;26(8):1196–1211. doi: [10.1002/bse.1978](https://doi.org/10.1002/bse.1978).
15. Qian W, Schaltegger S. Revisiting carbon disclosure and performance: legitimacy and management views. *British Account Rev.* 2017;49(4):365–379. doi: [10.1016/j.bar.2017.05.005](https://doi.org/10.1016/j.bar.2017.05.005).
16. Karim EA, Albitar K, Elmarzouky M. A novel measure of corporate carbon emission disclosure, the effect of capital expenditures and corporate governance. *J Environ Manage.* 2021;290:112581. doi: [10.1016/j.jenvman.2021.112581](https://doi.org/10.1016/j.jenvman.2021.112581).
17. IEA. Electrification; 2023. [Online]. Available from: <https://www.iea.org/energy-system/electricity/electrification>. [Accessed 30 Juli 2024]
18. Brander M, Gillenwater M, Asci F. Creative accounting: a critical perspective on the market-based method for reporting purchased electricity (Scope 2) emissions. *Energy Policy.* 2018;112:29–33. doi: [10.1016/j.enpol.2017.09.051](https://doi.org/10.1016/j.enpol.2017.09.051).
19. WBCSD and WRI. GHG protocol releases summary of Scope 2 guidance survey feedback; 2023. [Online]. Available from: https://ghgprotocol.org/sites/default/files/2023-11/Scope%202%20Survey%20Summary_Final.pdf.
20. Galzi P. Do green electricity consumers contribute to the increase in electricity generation capacity from renewable energy sources? Evidence from France. *Energy Policy.* 2023;179:113627. doi: [10.1016/j.enpol.2023.113627](https://doi.org/10.1016/j.enpol.2023.113627).
21. Bjørn A, Tilsted JP, Addas A, et al. Can science-based targets make the private sector Paris-aligned? A review of the emerging evidence. *Curr Clim Change Rep.* 2022;8(2):53–69. doi: [10.1007/s40641-022-00182-w](https://doi.org/10.1007/s40641-022-00182-w).

22. Science-based Targets Initiative. Companies taking action; 2024. [Online]. Available from: <https://science-basedtargets.org/companies-taking-action>.
23. IEA. 2024. Chemicals [Online]. Available from: <https://www.iea.org/energy-system/industry/chemicals>.
24. Bauer F, Tilsted JP, Pfister S, et al. Mapping GHG emissions and prospects for renewable energy in the chemical industry. *Curr Opin Chem Eng.* 2023;39: 100881. doi: 10.1016/j.coche.2022.100881.
25. Mulder M, Zomer SP. Contribution of green labels in electricity retail markets to fostering renewable energy. *Energy Policy.* 2016;99:100–109. doi: 10.1016/j.enpol.2016.09.040.
26. Paris A, Hechelmann RH, Buchenau N. Exploring the effect of guarantees of origin on the decarbonization of corporate electricity procurement: a case study of Germany and Norway. *J Indus Ecol.* 2024;28(6):1657–1669. doi: 10.1111/jiec.13553.
27. Busch T, Johnson M, Pioch T. Corporate carbon performance data: quo vadis? *J Indus Ecol.* 2022;26(1): 350–363. doi: 10.1111/jiec.13008.
28. SINARMAS CEP SA DEUTSCHLAND GMBH, 2022. 2022 Annual Sustainability Report [Online]; [cited 2024 Dec 10]. Available from: <https://sinarmascepsa.com/wp-content/uploads/sustainability-report/2022%20Annual%20Sustainability%20Report.pdf>
29. Yara, Integrated Report, 2022. [Online]; [cited 2024 Dec 10]. Available from: <https://www.yara.com/siteassets/investors/057-reports-and-presentations/annual-reports/2022/yara-integrated-report-2022.pdf>
30. Bachem Holding AG, 2023. Annual Report. [Online]; [cited 2024 Dec 10]. Available from: <https://www.bachem.com/about-bachem/investors-and-media/reports-and-presentations/>
31. Hempel, 2020. Annual Report 2020. [Online]; [cited 2024 Dec 10]. Available from: [hempel.com/-/media/files/global/pdf/news/2021/81497-hempel-annual-report-2020_web.pdf](https://www.hempel.com/-/media/files/global/pdf/news/2021/81497-hempel-annual-report-2020_web.pdf)
32. IPSEN, 2023. UNIVERSAL REGISTRATION DOCUMENT 2023. [Online]; [cited 2024 Dec 10]. Available from: https://www.ipsen.com/websites/ipsen_com_v2/wp-content/uploads/2024/04/17154251/EN_UNIVERSAL-REGISTRATION-DOCUMENT-2023-BD.pdf
33. OCI Global, 2023. Annual Report 2023. [Online]; [cited 2024 Dec 10]. Available from: https://investors.oci-global.com/sites/default/files/2024-04/OCI-Annual-Report-2023-vf_0.pdf
34. Lenzig Group, 2023. Annual and Sustainability Report 2023. [Online]; [cited 2024 Dec 10]. Available from: https://reports.lenzing.com/annual-and-sustainability-report/2023/_assets/downloads/entire-lenzig-ar23.pdf
35. Beiersdorf, 2023. Annual report 2023. [Online]; [cited 2024 Dec 10]. Available from: <https://reports.beiersdorf.com/annual-report/2023/>
36. Beiersdorf, 2023. GRI Content Index 2023. [Online]; [cited 2024 Dec 10]. Available from: <https://www.beiersdorf.com/sustainability/reporting/sustainability-reporting>
37. Sonova Holding AG, ESG Report 2021/22. [Online]; [cited 2024 Dec 10]. Available from: https://www.sonova.com/sites/default/files/2022-05/10_Sonova_AR_2021_22_ESG_report_en.pdf
38. Qiagen N.V., 2023. Sustainability Statement 2023. [Online]; [cited 2024 Dec 10]. Available from: https://s28.q4cdn.com/125951340/files/doc_financials/2023/ar/2023-QIAGEN-Sustainability-Statement-NFS.pdf
39. ICON PLC, 2023. ICON CARES 2023 Report. [Online]; [cited 2024 Dec 10]. Available from: https://www2.iconplc.com/2023_ESG_Report
40. Symrise, 2023. Sustainability Record 2023. [Online] Available at: https://symrise.com/de/unternehmensbericht/2023/downloads/SYM_gri_2023_EN.pdf [Accessed 10 December 2024]
41. Borregaard, 2023. Annual Report 2023; [cited 2024 Dec 10]. Available from: <https://www.borregaard.com/hubfs/Media/InvestorFiles/GeneralMeeting2024/annual-report-2023.pdf>
42. ZEISS Group, 2023. Sustainability Report 2022/2023. [Online]; [cited 2024 Dec 10]. Available from: <https://www.zeiss.com/corporate/en/about-zeiss/present/sustainability/sustainability-report.html#downloads>
43. IEA. Executive summary; 2022. Norway 2022. [Online]. Available from: <https://www.iea.org/reports/norway-2022/executive-summary>.
44. AIB. Activity statistics; 2023. [Online]. Available from: <https://www.aib-net.org/facts/market-information/statistics/activity-statistics-all-aib-members>.
45. AIB. European residual mixes, results of the calculation of residual mixes for the calendar year 2021, version. 2022. Available from: https://www.aib-net.org/sites/default/files/assets/facts/residual-mix/2023/AIB_2023_Residual_Mix_FINALResults09072024.pdf
46. Lanxess, 2020. Annual Report 2020. [Online]; [cited 2024 Dec 15]. Available from: https://www.annualreports.com/HostedData/AnnualReportArchive/l/lanxess_2020.pdf
47. L'Oréal, 2023. Universal Registration Document [Online] ; [cited 2024 Dec 15]. Available from: https://www.loreal-finance.com/system/files/2024-03/LOREAL_2023_Universal_Registration_Document_en.pdf
48. Synthomer, 2020. Annual Report [Online]; [cited 2024 Dec 15]. Available from: https://www.synthomer.com/media/3jchkojy/synthomer_ar2020.pdf
49. Treatt PLC. Annual report 2022; 2022. [Online]. Available from: <https://www.treatt.com/investor-relations/financial-results-presentations/annual-reports/treatt-annual-report-2022>.
50. Tecan Group. Annual report 2023; 2024. [Online]. Available from: <https://www.tecanannualreport.com/annual-report-2023>.
51. Guerbet. Universal registration document 2023; 2023. [Online]. Available from: <https://www.guerbet.com/investors/presentations-reports-regulated-information/>
52. Stratec SE. Financial reports; 2024. [Online]. Available from: <https://ir.stratec.com/websites/stratec/English/4000/financial-reports.html>
53. Stratec SE, 2021. Financial Reports. [Online]; [cited 2024 Oct 10]. Available from: <https://ir.stratec.com/websites/stratec/English/4000/financial-reports.html>
54. Stratec SE, 2022. Financial Reports. [Online]; [cited 2024 Oct 10]. Available from: <https://ir.stratec.com/websites/stratec/English/4000/financial-reports.html>

55. Stratec SE, 2023. Financial Reports. [Online]; [cited 2024 Oct 10]. Available from: <https://ir.stratec.com/websites/stratec/English/4000/financial-reports.html>
56. Stratec SE, 2024. Financial Reports. [Online]; [cited 2024 Oct 10]. Available from: <https://ir.stratec.com/websites/stratec/English/4000/financial-reports.html>
57. Global Reporting Initiative. GRI Standards English Language; 2023. [Online]. Available from: GRI – GRI Standards English Language.
58. U.S. Securities and Exchange Commission. The enhancement and standardization of climate-related disclosures for investors; 2024. [Online]. Available from: <https://www.sec.gov/rule-release/33-11275>.
59. IFRS. IFRS S2 climate-related disclosures; 2024. Available from: <https://www.ifrs.org/issued-standards/ifrs-sustainability-standards-navigator/ifrs-s2-climate-related-disclosures/>.

Appendix: List of companies included in the sample

Air Liquide, Alk-Abello A/S, Almirall S.A., AMG Advanced Metallurgical Group NV, ARKEMA, AstraZeneca, BACHEM HOLDING AG, BASF SE, Bayer AG, Beiersdorf AG, BERNER, Borregaard ASA, CARL ZEISS AG, Chiesi, Chr. Hansen

Holding A/S, Clariant AG, CLINIGEN GROUP PLC, COLOPLAST A/S, CONVATEC GROUP PLC, Covestro AG, Croda International, Dechra Pharmaceuticals PLC, DYNASOL ELASTOMEROS SAU, ELEKTA AB, Elementis plc, Elkem ASA, EMBLA MEDICAL HF, ERCROS SA, Essity, Evonik Industries AG, FIRMENICH SA, GENMAB A/S, Givaudan SA, GlaxoSmithKline, GRIFOLS, GRUPO FERRER INTERNACIONAL S.A., GUERBET, H&R Group, HEMPEL A/S, Henkel AG & Co. KGaA, Hikma Pharmaceuticals, Icon PLC, Indivior, Ipsen, Johnson Matthey, Kemira Corporation, Koninklijke DSM, LANXESS AG, Lenzing AG, L'Oréal, Lundbeck A/S, MANE, MCBRIDE PLC, MEDINCELL, MEDTRONIC PUBLIC LIMITED COMPANY, MERCK KGAA, MESSER SE & CO. KGAA, METTLER TOLEDO INTERNATIONAL INC, NOBIAN HOLDING 2 B.V., Nouryon, Novartis, Novo Nordisk A/S, Novozymes A/S, OCI N.V., OLEON HOLDING, Oriflame Holding Ltd; Orion Oyj, PCC Exol, Perrigo Company plc; Puig, S.L., PZ Cussons, QIAGEN N.V.; Recipharm Ab, Reckitt Benckiser, Recordati SpA, Robertet SA; SANOFI, SGL CARBON SE, Sika Group AG, SINARMAS CEPESA DEUTSCHLAND GMBH; SMITH & NEPHEW PLC, SNF SA, SOL SPA; Solvay S.A., SONOVA HOLDING AG, STRATEC SE, STRAUMANN HOLDING AG, Swedish Orphan Biovitrum, Symrise AG, Synthomer plc, TECAN GROUP AG, Tessengerlo Group, TOPSOE A/S, Treatt PLC, UCB SA, Unilever plc, Urenco Group, VICTREX PLC, VYGON, Wacker Chemie AG, Yara International ASA.